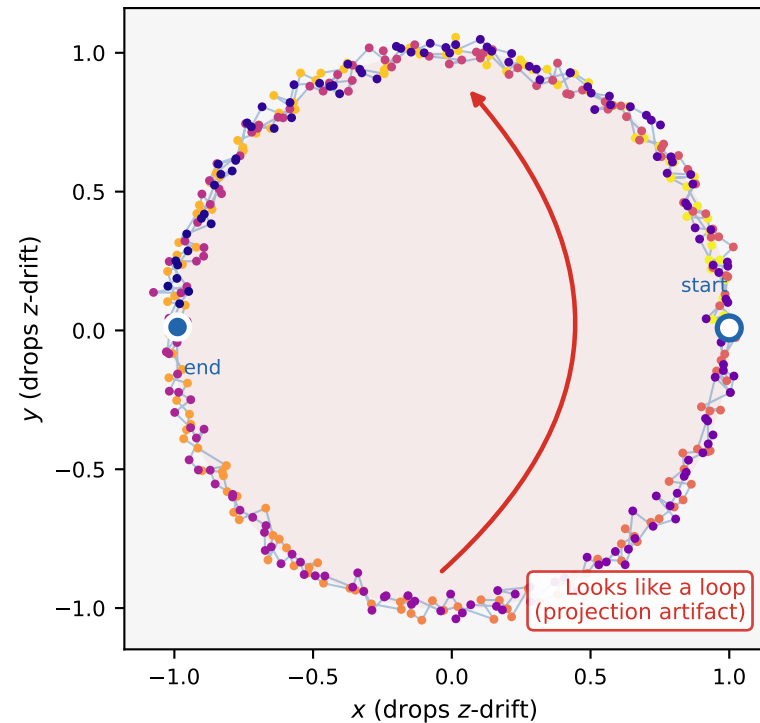
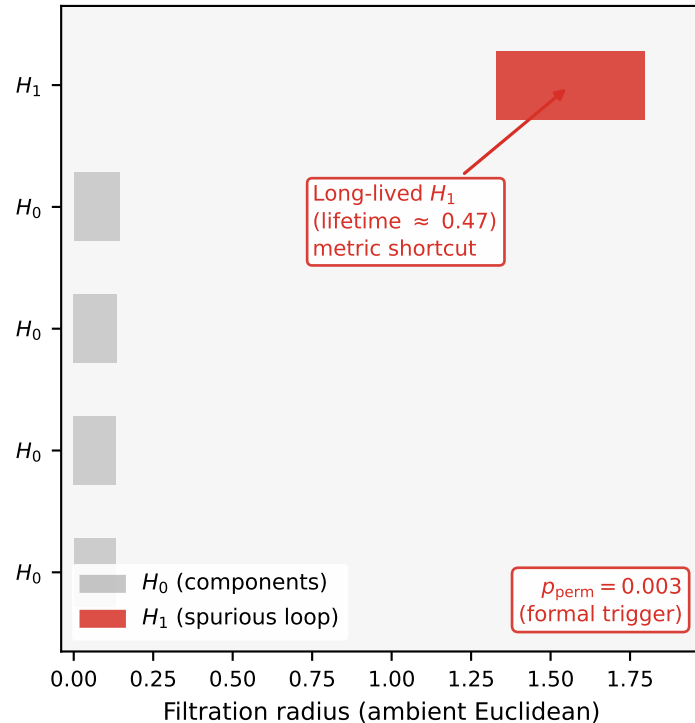


Why visual loops and Euclidean PH are not reliable recurrence evidence

(a) Drop-z projection of 3D helix
(contractible path looks circular)



(b) Ambient 3D Euclidean PH
(metric-shortcut false positive)



(c) Graph-geodesic PH
(metric shortcut suppressed)

